

$$\begin{aligned}
& \gamma_e^{11i2, i2} \rightarrow \\
& \frac{1}{16\pi^2} \left(-\frac{1}{2} \overline{C_{\phi 1\phi 2}} y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{3,1,0} [m_{\tilde{l}}^{pr}, m_{\tilde{e}}^{pr}] + \frac{3}{2} \overline{C_{\phi 1\phi 2}} y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{4,1,-1} [m_{\tilde{l}}^{pr}, m_{\tilde{e}}^{pr}] - \right. \\
& \overline{C_{\phi 1\phi 2}} y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{5,1,-2} [m_{\tilde{l}}^{pr}, m_{\tilde{e}}^{pr}] - \frac{3}{2} \overline{C_{\phi 1\phi 2}} y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{3,1,0} [m_{\tilde{q}}^{pr}, m_{\tilde{d}}^{pr}] + \\
& \frac{9}{2} \overline{C_{\phi 1\phi 2}} y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{4,1,-1} [m_{\tilde{q}}^{pr}, m_{\tilde{d}}^{pr}] - 3 \overline{C_{\phi 1\phi 2}} y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{5,1,-2} [m_{\tilde{q}}^{pr}, m_{\tilde{d}}^{pr}] - \\
& \frac{3}{2} \overline{C_{\phi 1\phi 2}} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} \text{LF}_{3,1,0} [m_{\tilde{u}}^{pr}, m_{\tilde{q}}^{pr}] + \frac{9}{2} \overline{C_{\phi 1\phi 2}} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} \text{LF}_{4,1,-1} [m_{\tilde{u}}^{pr}, m_{\tilde{q}}^{pr}] - \\
& 3 \overline{C_{\phi 1\phi 2}} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} \text{LF}_{5,1,-2} [m_{\tilde{u}}^{pr}, m_{\tilde{q}}^{pr}] + \overline{C_{\phi 1\phi 2}} g_1^2 y_e^{i1i2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_1] - \\
& 2 \overline{C_{\phi 1\phi 2}} g_1^2 y_e^{i1i2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] + \overline{C_{\phi 1\phi 2}} g_1^2 y_e^{i1i2} \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] + \\
& 3 \overline{C_{\phi 1\phi 2}} g_2^2 y_e^{i1i2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_2] - 6 \overline{C_{\phi 1\phi 2}} g_2^2 y_e^{i1i2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] + \\
& 3 \overline{C_{\phi 1\phi 2}} g_2^2 y_e^{i1i2} \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] - m_1 \tilde{\mu} g_1^2 y_e^{i1i2} \text{LF}_{1,1,1,0} [m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] + \\
& \frac{1}{4} m_1 g_1^2 (\overline{C_{\phi 1\phi 2}} a_e^{i1i2} + C_{\phi 1^2} \tilde{\mu} y_e^{i1i2}) \text{LF}_{2,2,1,-1} [m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] + m_1 \tilde{\mu} g_1^2 y_e^{i1i2} \\
& \text{LF}_{1,1,1,0} [m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] + \frac{1}{4} m_1 g_1^2 (\overline{C_{\phi 1\phi 2}} a_e^{i1i2} + C_{\phi 1^2} \tilde{\mu} y_e^{i1i2}) \text{LF}_{2,2,1,-1} [m_1, m_{\tilde{l}}^{i1}, m_{\tilde{e}}^{i2}] - \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^2 y_e^{i1i2} \text{LF}_{1,1,1,0} [m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{2} \overline{C_{\phi 1\phi 2}} g_1^2 y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] + \\
& \frac{1}{2} m_1 C_{\phi 1^2} \tilde{\mu} g_1^2 y_e^{i1i2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] + \frac{1}{4} \overline{C_{\phi 1\phi 2}} g_1^2 y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] - \\
& \frac{1}{4} m_1 C_{\phi 1^2} \tilde{\mu} g_1^2 y_e^{i1i2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_e^{i1i2} \text{LF}_{1,1,1,0} [m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
& \frac{3}{4} \overline{C_{\phi 1\phi 2}} g_2^2 y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{3}{4} m_2 C_{\phi 1^2} \tilde{\mu} g_2^2 y_e^{i1i2} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] - \\
& \frac{1}{2} m_1 g_1^2 (\overline{C_{\phi 1\phi 2}} a_e^{i1i2} + C_{\phi 1^2} \tilde{\mu} y_e^{i1i2}) \text{LF}_{2,1,1,0} [m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] + \\
& \frac{1}{2} m_1 g_1^2 (\overline{C_{\phi 1\phi 2}} a_e^{i1i2} + C_{\phi 1^2} \tilde{\mu} y_e^{i1i2}) \text{LF}_{3,1,1,-1} [m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] - \\
& \frac{1}{2} m_1 g_1^2 (\overline{C_{\phi 1\phi 2}} a_e^{i1i2} + C_{\phi 1^2} \tilde{\mu} y_e^{i1i2}) \text{LF}_{2,1,1,0} [m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}] + \\
& \left. \frac{1}{2} m_1 g_1^2 (\overline{C_{\phi 1\phi 2}} a_e^{i1i2} + C_{\phi 1^2} \tilde{\mu} y_e^{i1i2}) \text{LF}_{3,1,1,-1} [m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}] \right)
\end{aligned}$$